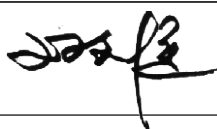
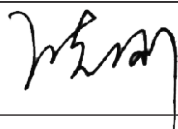
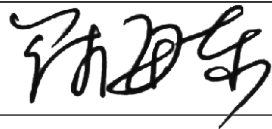
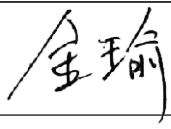


院教学指导委员会主任 (院长)	学院分管教学 (副院长)	审核人 (专业责任教授负责人)	执笔人
			

中国武汉科技大学与英国伯明翰大学合作举办 网络工程专业本科教育项目培养方案

**Academic Plan and Curriculum for Undergraduate
Network Engineering Program
Jointly Offered by Wuhan University of Science and Technology,
PRC and Birmingham City University, UK**

项目介绍

经教育部批准，武汉科技大学与英国伯明翰城市大学（Birmingham City University）合作举办网络工程专业本科项目，项目批准书编号为MOE42UK2A20141630N，办学层次和类别为本科学历教育，纳入国家普通高等教育招生计划。

本项目由中英双方共同制定人才培养方案和教学计划，并共同选派优秀教师任教，主要课程采用英语或双语教学，其中英方承担约1/3核心课程的教学。修业年限为4年，学生可在武汉科技大学完成全部学业，修完规定课程学分并考核合格，可获得武汉科技大学本科毕业证书和学士学位。

Background of Program

Upon approval of the Ministry of Education, PRC (Program Approval No. MOE42US2A20121259N) and pursuant to the Co-operative Education Agreement signed between the parties, Wuhan University of Science and Technology, China and Birmingham City University, UK have agreed to jointly offer the Bachelor of Engineering in Network Engineering Program beginning from 2015 for 4 years. The entry requirements of the Program are: (a) successful completion of the three-year senior secondary education in China, and (b) meeting the admissions requirements of WUST in the national College Entrance Examination.

The Program operates based on an Academic Plan and Curriculum jointly designed by both universities, which also provide qualified and experienced faculty to deliver teaching in the joint program up to due share (WUST 2/3 vs. BCU 1/3 for core courses). The medium of instruction of the courses will be English or English complemented by Chinese on the part of the WUST faculty.

一、培养目标

本专业培养思想素质高、基础扎实、实践能力强、具备德、智、体、美全面发展、具有国际视野和创新精神、能适应计算机网络科学与工程发展需要的高素质应用型人才。该专业毕业生能在网络应用软件开发、网络工程设计与规划、网络安全防护、网络运行维护与管理等领域发挥所学专长，逐渐成长为中坚骨干。期待毕业生五年左右达到以下目标：

1、具备扎实的数学、自然科学基础知识及其运用能力，具有良好的人文素养、社会责任感和职业道德；

2、掌握网络工程专业领域的基础理论和专业知识，具有创新意识、复杂网络系统建模、分析、设计、开发、应用与维护、技术管理等方面的工程实践能力；

- 3、具有文献检索和资料查询的能力，了解计算机网络领域的前沿动态和发展趋势；
- 4、具有良好的表达和沟通能力以及团队合作和组织管理能力；
- 5、具有终身学习的意识，具备不断学习和适应发展的能力；
- 6、具有良好的外语应用能力。

I.Training objectives

This major trains undergraduate students to be high-quality application-oriented talents with high ideological quality, solid foundation, practical ability, multi-skill technology, international vision and innovative spirit who are adapting the development needs of science and technology of computer network. The graduates of this major can utilize their specialities and gradually become the skeleton staff in the fields such as network software developing, network engineering designing and planning, network security protection and network maintenance and management. The graduated students are expected to reach the following goals in about five years:

1. Having solid knowledge of mathematics and natural science and mastering how to use them, and having good humanistic quality, social responsibility and professional ethics;
2. Mastering fundamental theory and professional knowledge in the field of network engineering, and having innovative consciousness and the ability of engineering practice of complex network system modeling, analysis, design, development, application, maintenance and technical management etc.;
3. Having the ability to literature retrieval and data query, and understanding the frontiers and trends in the field of computer network;
4. Having good skills in presentation, communication, teamwork and organizational management;
5. Having the consciousness of lifelong learning and the ability to keep on learning and adapting to development;
6. Having good ability to use foreign language.

二、毕业要求

- 1、工程基础知识与人文社科素养：掌握从事网络工程专业所需的相关数学、自然科学基础知识、工程技术知识和一定的人文社科知识。
- 2、问题分析：能够应用数学、自然科学和工程科学的基本原理，识别、表达、并通过文献研究分析网络系统需求、目标等专业领域复杂工程问题，以获得有效结论。
- 3、设计/开发解决方案：能够设计针对网络工程专业领域复杂工程问题的解决方案，设计满足特定需求的网络系统和装置、网络系统软件等方案或技术，并能够在设计环节中体现创新意识，考虑社会、健康、安全、法律、文化以及环境等因素。
- 4、实验设计与信息处理：能够基于科学原理并采用科学方法对网络系统数据处理、性能评估及修正改进等复杂工程问题进行研究，包括设计实验、分析与解释数据、并通过信息综合得到合理有效的结论。
- 5、现代工具的应用：能够针对网络工程专业领域的复杂工程问题，开发、选择与使用恰当的计算机技术、资源、现代工程工具和信息技术工具，包括对网络系统复杂工程问题的预测与模拟，并能够理解其局限性。
- 6、工程师社会责任意识：能够基于工程相关背景知识进行合理分析，利用网络工程设计、实施及评估规范评价网络工程专业工程实践和复杂工程问题解决方案对社会、健康、安全、法律以及文化的影响，并理解应承担的责任。
- 7、环境和可持续发展：能够理解和评价针对网络工程技术中复杂工程问题的工程实践对环境、社会

可持续发展的影响。

8、职业道德与规范：具有人文社会科学素养、社会责任感，能够在网络工程项目实践中理解并遵守工程职业道德和规范，履行责任。

9、团队合作：能够在多学科背景下的团队中承担个体、团队成员以及负责人的角色。

10、沟通：能够就网络系统设计、研究、开发等的复杂工程问题与业界同行及社会公众进行有效沟通和交流，包括撰写报告和设计文稿、陈述发言、清晰表达或回应指令，并具备一定的国际视野，能够在跨文化背景下进行沟通和交流。

11、项目管理：面向网络工程项目的多学科环境，理解、掌握并应用工程管理原理与经济决策方法。

12、终身学习：具有自主学习和终身学习的意识，有不断学习和适应发展的能力。

II. Requirements

1. Fundamental knowledge of engineering and basic humanistic and social science quality: mastering required related mathematics, fundamental knowledge of natural science, engineering knowledge and certain humanistic and social science quality which are necessary for network engineering major.

2. Problem analysis: having the ability to use the fundamental principles of mathematics, natural science and engineering to recognize express and analyze the network system requirements and targets for the engineering problems in professional area through the literature analysis and research and obtain the valid conclusion.

3. Design/Development solutions: having the ability to design and develop the solutions for the complex engineering requirements in the field of network engineering, design and develop solutions or technology for network systems and devices, system software with special requirements, and being able to reflect the innovative spirit in design with consideration on the social, health, safety, legal, cultural and environmental factors.

4. Experiment design and information processing: having the ability to research on the complex engineering problems in data processing of network systems, performance evaluation and modification/optimization, including experiment design, data analysis and interpretation, and the reasonable and effective conclusions through comprehensive information processing.

5. Application of modern tools: having the ability to develop, select and use appropriate computing technology, resources, modern engineering tools and information tools for the complex engineering problems in the field of network engineering including the prediction and simulation of the complex engineering problems in network systems and understand the their limitations.

6. Engineer's social responsibility consciousness: Having the ability to get the reasonable analysis based on engineering related background knowledge to evaluate the impacts of the network engineering practice and solutions for the complex engineering problems on the society, health, safety, law and culture by using the criteria of network engineering design, implementation and evaluation, and understand the corresponding responsibility.

7. Environments and sustainable development: having the ability to understand and evaluate the impacts on the environments and society from the engineering practice of the complex engineering problems in network engineering.

8. Professional ethics and norms: having the humanistic, social and scientific literacy and social responsibility, and understanding and abiding the engineering ethics and norms in network engineering projects and fulfilling their responsibilities.

9. Team work: having the ability to undertake the roles of individuals, team members or leaders in a multidisciplinary team.

10. Communication: having the ability to communicate effectively with the industry peers and the public for the complex engineering problems of the design, research and development of network system, including report writing, design stating, expressing clearly and responding the instructions, and communicate and exchange in the culture-crossing background with a certain international vision.

11. Project management: understanding, mastering and applying the engineering management principles and economic decision making approaches in network engineering oriented projects under the multidisciplinary environments.

12. Lifelong Learning: having an international perspective and good ability to use foreign language.

附：培养目标实现矩阵

	培养目标 1	培养目标 2	培养目标 3	培养目标 4	培养目标 5	培养目标 6
毕业要求 1	√					
毕业要求 2		√	√			
毕业要求 3	√	√				
毕业要求 4		√				
毕业要求 5		√				
毕业要求 6	√					
毕业要求 7			√			
毕业要求 8	√					
毕业要求 9				√		
毕业要求 10				√		√
毕业要求 11		√		√		
毕业要求 12					√	

三、专业主干课程

计算机组成原理、操作系统、通信基础、网络应用和协议、以太网、网络交换技术、路由技术、网络管理、高级网络设计、网络模拟和建模

III. Main courses

Principles of Computer Composition, Operating System, Communication basics, Network applications and protocols, Ethernet, Network Switched Technology, Routing Technology, Network Management, Advanced Network Design, Network Simulation and Modeling

四、基本学制：四年

IV. Recommended length of the program: 4years

五、授予学位：工学学士（标题宋体4号，加粗，具体内容宋体4号）

修业年限为4年，学生可在武汉科技大学完成全部学业，修完规定课程学分并考核合格，可获得武汉科技大学本科毕业证书和工学学士学位。

若本项目学生英语水平和学业成绩达到武汉科技大学和英国伯明翰城市大学要求，可赴伯明翰城市

大学计算机网络或其他相近本科专业学习一段时间。修完伯明翰城市大学所规定的课程并达到规定要求，可获得伯明翰城市大学向其本专业学生所颁发的相同的学士学位，同时还可获得武汉科技大学所颁发的本专业本科毕业证书和工学学士学位。

V. Degree: Bachelor of Engineering

Students in the program may choose to study for 4 years at WUST for WUST's Bachelor of Engineering degree in Mechanical Engineering and a relevant Certificate of Graduation subject to satisfactory completion of the Program.

As an alternative, a student can also choose to transfer to Birmingham City University's undergraduate Network Engineering Program based in BCU after having completed 2 or 3 years of study at WUST on condition that s/he meets the English language requirement of BCU and other requirements that may be imposed by either university. And a Bachelor's degree in Engineering can be conferred by Birmingham City University (BCU) after the completion of the academic requirements from BCU.

六、毕业学分要求：175学分

课程类型	学分要求	课程类型	学分要求
1、通识教育平台课程	48	3、专业课程模块	58.5
必修课程	44	必修课程	54
选修课程	4	选修课程	4.5
2、学科基础平台课程	42.5		
必修课程	37	4、实践教学模块	19
选修课程	5.5	5、素质拓展模块	7

VI. Credits required for graduation: 175 credits

Type of courses	Academic credits	Type of courses	Academic credits
1.Courses of general education	48	3. Specialized Courses	58.5
Required courses	44	Core specialized courses	54
Elective courses	4	Elective courses	4.5
2. General disciplinary courses	42.5		
Required courses	37	4.Practicum and internship courses	19
Elective courses	5.5	5. Quality development courses	7

七、学时学分比例

1、必修选修学分比例

类别	学分（学时）	占总学分（学时）比例
必修	161/2576	92%
选修	14/224	8%

2、实践教学环节学分比例

类别		学分（学时）	占总学分（学时）比例
实践教学环节	实验教学学时	23/365	32%
	实践教学模块	19/416	
	素质拓展模块	7/112	

八、毕业要求实现矩阵

VIII. Graduation Realization Matrix

课程名称	网络工程专业毕业要求											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
思想道德修养与法律基础	√						√	√				
中国近现代史纲要	√						√	√				
马克思主义基本原理	√							√				
毛泽东思想和中国特色社会主义理论体系概论	√											
形势与政策	√	√										
军事理论与训练	√											
体育								√				
英语	√											
高等数学		√										
线性代数		√										
概率论及数理统计		√										
大学生心理健康教育	√							√				
职业生涯规划与就业指导	√							√				√
公益劳动	√											
研究方法学		√										
商业情景技术		√										
学习、研究方法和技巧		√										
程序设计基础			√									
程序设计实践				√	√							
计算机组成原理			√									
计算机组成原理实践				√								
Python 编程技术			√		√							
面向对象的分析和设计			√							√		
面向对象的程序设计方法		√	√									
操作系统			√									
通信基础		√	√									
以太网			√									
网络交换技术			√									
路由技术			√									

课程名称	网络工程专业毕业要求											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
系统集成			√		√							
网络管理				√	√	√	√					
高级网络设计			√							√		
网络模拟和建模				√	√							
计算技术创新与应用						√	√		√	√		√
计算机软硬件集成实习				√		√	√			√		
商业情景技术课程设计				√						√		
面向对象的程序设计方法 课程设计				√	√					√		
以太网课程设计				√	√					√		
面向对象的分析和设计课 程设计				√	√					√		
生产实习				√	√	√		√	√	√	√	
毕业实习				√	√	√		√	√	√	√	
毕业设计(论文)			√	√	√	√	√	√		√		√

体育、军事理论、马克思主义基本原理等思想政治课



十、教学环节设置及学分分布表

(四年制本科 2018级 软件工程“产业计划”专业)

X、Offered Course and Distribution of Academic Credits

课 程 类 型	课程 性质	课 程 编 码	课 程 名 称	学 分	合 计	课 内 学 时			实 践 学 时	建 议 学 期	先 修 课 程/备 注
						讲 课	实 验	上 机			
平 台	必修	1501882	体育(一) Physical Education(I)	1	26	26	0	0	0	1	
		1501883	体育(二) Physical Education(II)	1	34	34	0	0	0	2	
		2199059	中级英语(一) Intermediate English I	6	96	96	0	0	0	1	
		2199060	中级英语(二) Intermediate English II	6	96	96	0	0	0	2	
		2199061	高级英语(一) Advanced English I	4	64	64	0	0	0	3	
		2199062	高级英语(二) Advanced English II	4	64	64	0	0	0	4	
		2501001	军事理论与训练 Military Theory and Training	3	3周	0	0	0	3周	1	
		2501002	公益劳动 Community Service	1	16	0	0	0	16	4	
		2501003	职业生涯规划与就业指导 Career Planning and Employment Guidance	1	16	16	0	0	0	2	
		2501004	大学生心理健康教育 Mental Health Education	1	16	16	0	0	0	1	
		5101001	毛泽东思想与中国特色社会主义理论体系概论 Theoretical System of Socialism with Chinese Characteristics	5	80	64	0	0	16	4	
		5102001	马克思主义基本原理 Fundamentals of Marxism	3	48	44	0	0	4	3	
		5103001	中国近现代史纲要 An Outline of Modern and Contemporary History of China	3	48	42	0	0	6	2	
		5105001	思想道德修养与法律基础 Moral Cultivation and Basics of Law	3	48	42	0	0	6	1	
		5106001	形势与政策 World Affairs and State Policy	2	64	64	0	0	0	1, 2, 3, 4, 5, 6, 7, 8	
	选修		人文社科类1学分 Humanity and Social Science 1Academic Credits								

教学环节设置及学分分布表

(四年制本科 2018级 软件工程“产业计划”专业)

Offered Course and Distribution of Academic Credits

课 程 类 型		课程性质	课 程 编 码	课 程 名 称	学 分	合 计	课 内 学 时			实践学时	建议学期	先 修 课 程/备 注
							讲课	实验	上机			
平台	通识教育平台课程	选修		艺术体育类1学分 Artistic and Sports 1Academic Credits								
				自然科学类1学分 Natural Science 1Academic Credits								
				经济管理类1学分 Economic and Management 1Academic Credits								
	学科基础平台课程	必修	0702026	线性代数 Linear Algebra	2	32	32	0	0	0	3	
			0702303	概率论与数理统计A Probability and Mathematical Statistics (A)	3	48	48	0	0	0	4	
			0702601	高等数学A(一) Advanced Mathematics A(I)	5	80	80	0	0	0	1	
			0702602	高等数学A(二) Advanced Mathematics A(II)	6.5	104	104	0	0	0	2	
			1303836	程序设计基础(一) Basics of Programming (I)	3	48	48	0	0	0	1	
			1303837	程序设计基础(二) Basics of Programming (II)	3	48	48	0	0	0	2	
			1303841	程序设计实践(一) Practice Experience in Programming (I)	1	24	0	0	24	0	1	
			1303844	程序设计实践(二) Practice Experience in Programming(II)	1	24	0	0	24	0	2	
			2103014	商业情景技术 Context of Business	4.5	72	62	0	10	0	3	
			2103015	学习、研究方法和技巧 Learning, Study Methods and Skills	3	48	38	0	10	0	3	
			2103017	研究方法学 Research Methodology	5	80	80	0	0	0	3	
		选修	0702019	数学建模 Mathematical Modeling	2	32	28	0	4	0	5	
			1301220	计算机网络基础 Computer Network Basis	2	32	32	0	0	0	2	
			1302036	数据库系统原理 Principles of Database System	3.5	56	42	0	14	0	4	

教学环节设置及学分分布表

(四年制本科 2018级 软件工程“产业计划”专业)

Offered Course and Distribution of Academic Credits

课 程 类 型	课程 性质	课 程 编 码	课 程 名 称	学 分	合 计	课 内 学 时			实 践 学 时	建 议 学 期	先 修 课 程/备 注
						讲 课	实 验	上 机			
平 台	学科基 础平台 课程	选 修	1302083 人工智能导论 Introduction to Artificial Intelligence	2	32	32	0	0	0	3	
模 块	专业 课程 模块	必 修	1301026 计算机组成原理 Principles of Computer Composition	3.5	56	56	0	0	0	4	
			1302009 操作系统 Operating System	3	48	40	0	8	0	5	
			1302116 Python编程技术 Python Programming	5	80	66	0	14	0	5	
			1303002 计算机组成原理实践 Practice Experience in Principles of Computer Composition	1	16	0	16	0	0	4	
			1399003 计算技术创新与应用 Computing Technology Innovation and Application	1	16	16	0	0	0	7	
			2103001 面向对象的分析和设计 Object Oriented Analysis and Design	4	64	50	0	14	0	5	
			2103002 面向对象的程序设计方法 Object Oriented programming concepts	4	64	50	0	14	0	5	
			2103004 通信基础 Communication basics	5	80	66	14	0	0	4	
			2103006 以太网 Ethernet	4.5	72	58	14	0	0	4	
			2103008 网络交换技术 Network Switched Technology	5	80	66	14	0	0	6	
			2103009 路由技术 Routing Technology	3.5	56	42	14	0	0	6	
			2103010 系统集成 Integrated network technologies	3.5	56	42	14	0	0	6	
			2103011 网络管理 Network Management	5	80	66	14	0	0	7	
			2103012 高级网络设计 Advanced Network Design	3	48	34	14	0	0	7	
			2103013 网络模拟和建模 Network Simulation and Modeling	3	48	34	14	0	0	7	

教学环节设置及学分分布表

(四年制本科 2018级 软件工程“产业计划”专业)

Offered Course and Distribution of Academic Credits

课 程 类 型	课程 性质	课 程 编 码	课 程 名 称	学 分	合 计	课 内 学 时			实 践 学 时	建 议 学 期	先 修 课 程/备 注
						讲 课	实 验	上 机			
模 块	专业 课程 模块	选修	1301047 数字逻辑与数字系统 Digital Logic and Digital System	4	64	54	10	0	0	3	
			1301048 嵌入式系统设计与开发 Design and Development of Embedded System	2.5	40	30	10	0	0	6	
			1301066 专业英语 Specialized English	2	32	32	0	0	0	5	
			1302031 软件工程 Software Engineering	2.5	40	32	0	8	0	7	
			1302066 Linux内核与程序设计 Linux Kernel and Programming	2.5	40	30	0	10	0	6	
			1302105 无线网络与移动计算技术 Wireless Network and Mobile Computing	2.5	40	30	0	10	0	6	
			1302106 J2EE架构 J2EE Architecture	2.5	40	30	0	10	0	6	
			1303029 离散数学 Discrete Mathematics	4	64	64	0	0	0	2	
			1303079 算法设计与分析 Algorithm Design and Analysis	3	48	38	0	10	0	4	
			1303849 软件定义网络 Software Define Network	2.5	40	30	0	10	0	6	
			1303850 分布式计算 Distributed Computing	2.5	40	30	0	10	0	6	
	实践教学 模块	必修	1301065 生产实习 Production Practice	3	3周	0	0	0	3周	7	
			1301081 计算机软硬件集成实习 Fieldwork in Computer Software and Hardware Integration	1	1周	0	0	0	1周	4	
			1302097 毕业实习 Pre-graduation Internship	3	3周	0	0	0	3周	8	
			1302099 毕业设计(论文) Undergraduate Project (Thesis)	8	13周	0	0	0	13周	8	
			1302118 以太网课程设计 Course Project in Ethernet	1	1.5周	0	0	0	1.5周	5	

教学环节设置及学分分布表

(四年制本科 2018级 网络工程“国际”专业)

Offered Course and Distribution of Academic Credits

课 程 类 型	课程 性质	课 程 编 码	课 程 名 称	学 分	合 计	课 内 学 时			实 践 学 时	建 议 学 期	先 修 课 程/备 注
						讲 课	实 验	上 机			
模 块	实 践 教 学 模 块	必 修	1302119 面向对象的分析和设计课程设计 Course Project in Object Oriented Analysis and Design	1	1.5周	0	0	0	1.5周	6	
			1302120 面向对象的程序设计方法课程设计 Course Project in Object Oriented Programming Concepts	1	1.5周	0	0	0	1.5周	6	
			1302121 商业情景技术课程设计 Course Project in Context of Business Technology	1	1.5周	0	0	0	1.5周	4	
	素 质 展 拓 模 块	必 修	创新创业教育 Innovation Course1 Academic Credits								
			创新创业教育 Innovative Practices 2 Academic Credits								
			第二课堂 Second Classroom 3 Academic Credits								
			心理健康 Practices of Mental Health Education 1 Academic Credits								